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GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: POWER STEERING CONTROL MODULE (PSCM)

(G2027765)

DESCRIPTION AND OPERATION

POWER STEERING CONTROL MODULE (PSCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.
- BEFORE starting any repairs or investigations of power steering issues, first download and retain the SDD session file using the manufacturer-approved diagnostic system. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms.
- During normal vehicle operation steering assistance may be temporarily reduced in a controlled way under some circumstances. When diagnosing potential steering assistance issues, first check if the customer reported symptom relates **either** to one of these circumstances (as described in the relevant Description and Operation section in the workshop manual, see Section 211), **or**, to one of the following circumstances: an over temperature condition due to exceptionally demanding steering inputs; the vehicle power supply being in a low state of charge; CAN network signals required from other vehicle systems being unavailable or invalid

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Power Steering Control Module (PSCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: Power Steering (211-02 Power Steering, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1304-04	Electronic Power Assisted Steering System - System Internal Failures	**NOTE:** This DTC will log with the ignition set to ON ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Steering gear internal failure	**NOTE:** Once the repair is complete, clear DTC's, close the diagnostic session and then upload the session file to JLR. Failure to follow this instruction may result in a warranty claim becoming invalid ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest ▪ Check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms. If the time of DTC occurrence does not relate to the customer reported symptom, check for other appropriately timed and relevant DTCs in all vehicle modules. If the outcome of these investigations is not conclusive, contact the JLR Technical Helpdesk ▪ If the steering gear fault persists, check and install a new steering gear as required
B1304-07	Electronic Power Assisted Steering System - Mechanical failures	▪ Tire pressure(s) incorrect ▪ Wheel/tire size incorrect ▪ Suspension/steering damaged ▪ Power steering control module is not configured correctly	▪ Check and adjust the tire pressures as necessary ▪ Check the wheels and tires comply with the manufacturer's specification for the vehicle ▪ Check the suspension and steering for damage ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software
B1304-16	Electronic Power Assisted Steering System - Circuit voltage below threshold	**NOTE:** This DTC will log with the ignition set to ON ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	**NOTE:** The system is designed to give a heavier but stable steering feel when a low voltage is detected. It is recommended to reduce drastic inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate, high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ If the fault persists and no battery or charging issues can be identified, contact the JLR Technical Helpdesk. Do not replace the steering gear

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1304-17	Electronic Power Assisted Steering System - Circuit voltage above threshold	**NOTE:** This DTC will log with the ignition set to ON ▪ Battery/charging system fault	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ If the fault persists and no battery or charging issues can be identified, contact the JLR Technical Helpdesk. Do not replace the steering gear
B1304-1C	Electronic Power Assisted Steering System - Circuit voltage out of range	▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	**NOTE:** The system is designed to give a heavier but stable steering feel when a low voltage is detected. It is recommended to reduce drastic inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate, high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
B1304-98	Electronic Power Assisted Steering System - Component or system over temperature	▪ Power steering control module over temperature due to hot climate or excessive steering inputs by driver	**NOTE:** The system is designed to give a heavier but stable steering feel when high temperature is detected to protect the system from permanent damage. It is recommended to reduce level of steering inputs to the steering in the specific scenarios identified by the customer to provoke the problem. Depending on the level of reduction this could be accompanied by an alert ▪ Check for heat shield damage or mud packing around power steering control module. Allow the power steering control module to cool. Consider the environmental temperature. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C102D-07	High Friction Inside Power Steering - Mechanical failures	**NOTE:** This DTC will log with the ignition set to ON • Abnormally high friction in steering ball joints or suspension components • Abnormally high friction in steering gear	**NOTES:** • Do not damage bellows on steering gear removal. Failure to follow this instruction may result in warranty becoming invalid for steering gear assembly • DO NOT reset DTCs. Record the diagnostic session file and upload to JLR. Failure to follow this instruction may result in warranty claim becoming invalid • If this DTC is logged and BEFORE starting any repairs or further investigations of power steering issues, first download and retain the diagnostic session file using the Jaguar Land Rover Approved Diagnostic Equipment. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms • Disconnect the steering gear from the suspension. Check the suspension system for freedom of movement or abnormally high friction • Remove the steering gear boots and check for evidence of water inside the steering gear boots. If evidence of water ingress is present, replace the steering gear. Refer to the new module/component installation note at the top of the DTC index • Check the steering gear for abnormally high friction. If abnormally high friction is found, check if the onset of the steering system fault symptoms occurred at the same time that the DTC was logged. If the customer reports of heavy steering coincide with the logging of this DTC, then install new steering gear as required • If the DTC was logged between medium and high time (medium to high mileage) in service it is most likely caused by impact or water ingress damage. Please refer to - Electric Power Steering Gear Damage Assessment Checklist – In section 211-02 Diagnosis and Testing of the workshop manual
C200B-01	Steering Shaft Torque Sensor 1 - General electrical failure	**NOTE:** This DTC will log with the ignition set to ON • Power steering control module torque sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance • Power steering control module torque sensor internal failure	**NOTE:** DO NOT reset DTCs. Record the diagnostic session file and upload to JLR. Failure to follow this instruction may result in warranty claim becoming invalid • Refer to the electrical circuit diagrams and check the security and integrity of the power steering control module torque sensor circuit cable and connectors. Check for short circuit to ground, short circuit to power, open circuit, high resistance. Rectify as required • Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest. If the fault persists, check and install a new steering gear as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C200B-31	Steering Shaft Torque Sensor 1 - No signal	**NOTE:** This DTC will log with the ignition set to ON ▪ Power steering control module torque sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Power steering control module torque sensor internal failure	**NOTE:** DO NOT reset DTCs. Record the diagnostic session file and upload to JLR. Failure to follow this instruction may result in warranty claim becoming invalid ▪ Refer to the electrical circuit diagrams and check the security and integrity of the power steering control module torque sensor circuit cable and connectors. Check for short circuit to ground, short circuit to power, open circuit, high resistance. Rectify as required ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest. If the fault persists, check and install a new steering gear as required
C1B00-62	Steering Angle Sensor - Signal Compare Failure	▪ Wheel/tire size incorrect ▪ External received steering angle signal from steering angle sensor module is >90deg different than power steering control module internal measure of steering angle ▪ Steering angle sensor module incorrectly installed (installed when road wheels are not straight) ▪ Front road wheel alignment incorrect ▪ Steering angle sensor module fault	▪ Check the wheels and tires comply with the manufacturer's specification for the vehicle ▪ With the road wheels set straight ahead, check the steering wheel is straight. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Steering Wheel Angle (0x203A) = 0 degrees ▪ If the steering wheel angle is greater than 20 degrees out when the road wheels are straight ahead, check the installation of the steering angle sensor module and reinstall as necessary. Clear the DTC and retest ▪ Check the front road wheel alignment and rectify as necessary. Clear the DTC and retest ▪ Check the steering angle sensor module for any relevant DTCs. If fault(s) present, rectify as necessary ▪ If no fault found, clear the power steering control module DTC's and retest by conducting a static lock-to-lock steering maneuver (do not replace any components on the basis of one occurrence of this DTC)
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module via the high speed CAN bus (chassis)	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0300-56	Internal Control Module Software Incompatibility - Invalid/incomplete configuration	▪ Incorrect body control module installed ▪ Incorrect power steering control module installed	▪ Check for related DTCs stored in other modules. Check that the correct body control module is installed to the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the body control module with the latest level software ▪ Check that the correct power steering control module is installed to the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software. Perform routine - Power Steering Soft Lock End Stops Reset Routine
U1A4C-53	Build / End of Line Mode Active - Deactivated	**NOTE:** This DTC is for use during production only ▪ Power steering control module end of line is active	▪ Using the Jaguar Land Rover approved Diagnostic Equipment, run diagnostic routine - Set ECU in to Customer Mode (0x0E11) - Clear DTC and retest
U2001-00	Reduced System Function - No Sub Type Information	**NOTE:** This DTC will log with the ignition set to ON ▪ Power steering control module power or ground circuit open circuit, high resistance ▪ Power steering control module over temperature due to hot climate or excessive steering inputs by driver	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance ▪ Check for heat shield damage or mud packing around power steering control module. Allow the power steering control module to cool. Consider the environmental temperature. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest
U2001-07	Reduced System Function - Mechanical Failures	▪ Tire pressure(s) incorrect ▪ Wheel/tire size incorrect ▪ Suspension/steering damaged ▪ Power steering control module is not configured correctly	▪ Check and adjust the tire pressures as necessary ▪ Check the wheels and tires comply with the manufacturer's specification for the vehicle ▪ Check the suspension and steering for damage ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software
U2001-16	Reduced System Function - Circuit Voltage Below Threshold	▪ Battery/charging system fault ▪ Power steering control module power or ground circuit open circuit, high resistance	**NOTE:** The system is designed to give a heavier but stable steering feel when a low voltage is detected. It is recommended to reduce drastic inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate, high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2001-17	Reduced System Function - Circuit Voltage Above Threshold	**NOTE:** This DTC will log with the ignition set to ON • Power steering control module power or ground circuit open circuit, high resistance • Battery/charging system fault	**NOTE:** The system is designed to give a heavier but stable steering feel when a low voltage is detected. It is recommended to reduce drastic inputs to the steering in these scenarios, for example recently after start-up, extreme cold climate, high electrical system utilization/load. Depending on the level of reduction this could be accompanied by an alert • Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signals - Main ECU Supply Voltage (0xDD02) and Vehicle Battery Voltage (0x402A). Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits and associated connectors for open circuit, high resistance. Repair or replace circuit(s)/connector(s) as required. Clear DTC and retest • Refer to the relevant section of the workshop manual and test the battery and charging system
U2001-98	Reduced System Function - Component Or System Over Temperature	• Power steering control module over temperature due to hot climate or excessive steering inputs by driver	**NOTE:** The system is designed to give a heavier but stable steering feel when high temperature is detected to protect the system from permanent damage. It is recommended to reduce level of steering inputs to the steering in the specific scenarios identified by the customer to provoke the problem. Depending on the level of reduction this could be accompanied by an alert • Check for heat shield damage or mud packing around power steering control module. Allow the power steering control module to cool. Consider the environmental temperature. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest
U2100-56	Initial Configuration Not Complete - Invalid/incomplete configuration	• Power steering control module is not configured correctly	• Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software. Perform routine - Lock Steering Tune To Vehicle (3038)
U2300-54	Central Configuration - Missing calibration	• Car configuration file mismatch with vehicle specification • Power steering control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds, followed by another ignition cycle before clearing the DTCs • Using the Jaguar Land Rover Approved Diagnostic Equipment, check and up-date the car configuration file as necessary. Clear the DTC and re-test • Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2300-55	Central Configuration - Not configured	▪ Car configuration file mismatch with vehicle specification ▪ Power steering control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds, followed by another ignition cycle before clearing the DTCs ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check and up-date the car configuration file as necessary. Clear the DTC and re-test ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software
U2300-56	Central Configuration - Invalid/incomplete configuration	▪ Car configuration file mismatch with vehicle specification ▪ Power steering control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds, followed by another ignition cycle before clearing the DTCs ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check and up-date the car configuration file as necessary. Clear the DTC and re-test ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the power steering control module with the latest level software
U3001-00	Control Module Improper Shutdown - No sub type information	**NOTE:** This DTC may set when the vehicle battery is disconnected, whilst the ignition is in the ‘on’ position. This is not a fault. Clear the DTC and retest ▪ Occurrence of an unexpected power steering control module system reset	**NOTES:** ▪ In some case this can be induced during system reduction functions. Ignore if set in conjunction with other DTCs ▪ If this DTC is logged and BEFORE starting any repairs or further investigations of power steering issues, first download and retain the diagnostic session file using the Jaguar Land Rover Approved Diagnostic Equipment. Where possible, check if the timeframe of any stored DTCs coincides with customers' reports both of the onset of steering system fault symptoms and with the technical nature of the reported symptoms. Ignore this DTC unless customer complains of heavy or intermittent power steering assistance ▪ Refer to the electrical circuit diagrams and check the power steering control module power and ground circuits for open circuit, high resistance. Clear the DTCs and retest

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